

ROBOFEST- GUJARAT 6.0

Guidelines for submission of Ideation Proposal
Senior category

All teams must be present and ready to perform their challenge. The committee may assign any team to the task at random. Any kind of absence and lack of readiness will be subject to suitable actions, including the deduction of score/disqualification.

Students are required to perform their task and answer the questions. Mentors should not be involved in the task. The committee may allow mentors during question-answer if it feels necessary, depending on the situation.

Sample images given in these guidelines are only for visual understanding.

Application-Based Robot (ABR) Challenges

Game Overview:

The Application-Based Challenge is designed to simulate real-world problem-solving using robotics under specific societal, environmental, and industrial themes. Participants select one thematic track and design a robot to perform application-specific missions within a defined arena. This category prepares teams for participation on national and international platforms by encouraging innovation in socially impactful domains.

This category will be divided into two sub-categories, namely: **[1] ABR-I & [2] ABR-II**. ABR-I will have thematic tracks involving novel robots having rigid / semi-rigid arms. ABR-II will have the identical thematic tracks using novel robots made up with flexible / compliant / functionally graded/ soft materials.

Thematic Tracks & Game Descriptions:

NOTE: The below-mentioned Thematic Tracks will be common to both ABR-I & ABR-II.

a. Robots for Healthcare

Challenge: Building a robotic prototype for hospital service delivery.

Tasks:

- Navigate hospital corridors (marked paths).
- Detect and deliver medical items (dummy kits) to specific patient rooms.
- Avoid “infection zones” (marked danger zones).

Scoring:

- Navigation accuracy: 20 pts
- Successful delivery: 30 pts/item
- Avoiding infection zones: 20 pts

b. Robots for Humanitarian and Assistive Purpose

Challenge: Building a robotic prototype for disaster rescue and aid delivery.

Tasks:

- Navigate rubble-filled zone.
- Identify “victims” (heat or color-coded dummies).
- Deliver food/water packs.

Scoring:

- Victim identification: 20 pts/victim
- Aid delivery: 30 pts/pack
- Time bonus for complete mission under 5 mins: 20 pts

c. Robots for Educational/Research Purposes

Challenge: Building a robotic prototype for the demonstration of intelligent learning behavior.

Tasks:

- Complete a pattern-learning task (line/color/shape recognition).
- Adaptive response to changing tasks.

Scoring:

- Learning task accuracy: 40 pts
- Adaptation to new rule: 30 pts
- Presentation of algorithm: 30 pts (judged)

d. Robots for Community Services or Awareness

Challenge: Building a robotic prototype for public hygiene or waste segregation.

Tasks:

- Identify biodegradable vs. plastic items using vision sensors.
- Please Place them in correct bins.

Scoring:

- Correct classification: 20 pts/item
- Mission completion within time: 20 pts
- Creativity in awareness mechanism: 20 pts

e. Robots for Climate Change Challenges

Challenge: Building a robotic prototype for environment cleanup and monitoring.

Tasks:

- Detect pollutant patches (e.g., dark mats).
- Simulate collection via scoop/arm.
- Display pollution level on screen/speaker.

Scoring:

- Patch detection: 25 pts
- Accurate pickup: 25 pts
- Sensor data logging: 20 pts

You can select any one of the pollutants ~ be it in air, soil or water.

f. Robots for Sustainability

Challenge: Building a robotic prototype for smart agriculture.

Tasks:

- Line-following to simulate field navigation.
- Detect “crops” (green markers) and “weeds” (red markers).
- Spray or mark correctly.(non-toxic fluid)

Scoring:

- Crop/weeds classification: 30 pts
- Line-following accuracy: 20 pts
- Efficient spraying/marking: 30 pts

The Field will be real and navigation has to be through computer vision technique. You need to mark correctly as well as demonstrate adequate spraying by the robot.

Signs / markings can be on floor / field as well as on an elevated surface ~ 15 cm height.

g. Robots for Industrial Applications

Challenge: Building a robotic prototype for autonomous warehouse logistics or soft robotic systems for process plants.

Tasks:

- Pick, transport, and place items to designated storage points.
- Avoid dynamic obstacles (moving bot or object).

Scoring:

- Correct pickup & placement: 30 pts/item
- Route optimization: 20 pts
- Obstacle avoidance: 20 pts

The shape, dimensions & weight of the object depends on the application that you are aiming at. Ideally, the robot should be able to demonstrate its function for a payload of 5 kg (max.).

The ideal situation for 'pick-and-place' operation for your robot should be commensurate to industrial applications, e.g, material handling in process plants or manufacturing units. On an average, you can assume a height of 6-7 ft. above a suitable horizontal datum for the pick and place job.

h. Robots for World Peace

Challenge: Building a robotic prototype for surveillance and patrolling.

Tasks:

- Navigate perimeter path.
- Detect objects/colors representing "suspicious activity."
- Trigger alarm via sound/light.

Scoring:

- Whole patrol without error: 30 pts
- Accurate threat detection: 30 pts
- Alarm mechanism: 20 pts

Common Game Field Design Framework:

- Arena Size: Minimum 4m x 4m.
- Zone Types: Simulation zones vary based on theme—e.g., hospital wards, rubble, farmlands and cityscapes.
- Navigation Aids: Line tracks, colored zones, QR codes, dummy markers.
- Interactivity: Dummy props, sensors, simulated hazards, or AI-based interpretation zones.

NOTE: All teams under ABR-I and ABR-II categories will be provided with case-specific GUIDELINES for the GRAND FINALE, based on the hardware / working model shown at the PoC level. The above-mentioned challenges under headings 'a' to 'h' are just indicative of the basic requirements and performance that are expected from the teams. GUJCOST & TAC will have the full right to revise these general guidelines after evaluating the PoC models.

General Scoring Matrix:

Criteria	Points Range
Mission Task Completion	Up to 60
Technical Accuracy & Execution	Up to 20
Innovation/Creativity in Design	Up to 10
Time Efficiency	Up to 10
Penalties (rule violation, manual intervention)	-5 to -20
Maximum Score per team: 100 Points	
Minimum Qualifying Score: 60 Points	

Technical Requirements:

- Fully autonomous operation
- Use of vision, sensors, or AI (based on theme)
- Robots must fit within a 30cm x 30cm x 30cm bounding box

The 'bounding box' dimensions represent the most ideal situation of robotic structure in static condition. However, the robot can have a larger WORKSPACE that will certainly surpass the static dimensions of the 'bounding box'. Hence, the robot that you will design can surely extend its reach (e.g, beyond 30 cm).

- Optional presentation to judges for selected themes (educational, climate, peace)